

Notes: Koijen & Yogo (JPE, forthcoming)

Exchange Rates and Asset Prices in a Global Demand System

Contents

1	Big picture	3
2	Data and measurement (key objects)	3
2.1	International portfolio holdings	3
2.2	Asset prices and macro characteristics	4
2.3	What Table 1 and Table 2 show	4
2.4	Reduced-form price-quantity facts	4
3	Models and empirical specifications (core equations + interpretation)	5
3.1	Expected returns	5
3.2	Model of short-term rates	5
3.3	Nested logit demand system	5
3.4	Portfolio flows and market clearing	6
3.5	Estimating equation for demand within asset class	6
3.6	Reduced-supply identification example	7
3.7	Constructed instrument for inner-nest demand	7
3.8	Demand across asset classes	7
3.9	Aggregate demand elasticity and variance decomposition	8
4	Identification and instruments (what makes the estimates credible)	8
4.1	Inner nest: residual supply from geography and size	8
4.2	Outer nest: home bias as a shifter of outside weights	8
4.3	Why the design is compelling	9
4.4	Main caveats	9
5	Tables and figures	9
5.1	Table 1: Market values of financial assets	9
5.2	Table 2: Top ten investors by asset class	9
5.3	Figures 1–3: Relative prices versus quantities	10

5.4	Table 3: Predictive regressions	11
5.5	Figure 4: Illustration of residual supply	11
5.6	Table 4: Estimated demand within asset class	11
5.7	Table 5: Estimated demand across asset classes	13
5.8	Table 6: Estimated model of short-term rates	14
5.9	Table 7: Variance decomposition of exchange rates and asset prices	14
5.10	Table 8 and Figure 6: The European sovereign debt crisis	14
5.11	Table 9: Mean latent demand	15
5.12	Table 10 and Figure 7: Convenience yield on US assets	15
6	Short summary	17
7	Conclusion	17
7.1	Core conclusion (what the paper establishes)	17
7.2	Economic intuition	17
7.3	Takeaway	18

1 Big picture

- **Question.** How do international portfolio holdings and portfolio flows determine exchange rates, short-term rates, long-term yields, and equity prices across countries?
- **Core friction.** Standard international macro models do not fit the gravity effect and home bias in actual international portfolios. The paper replaces optimal frictionless portfolios with directly estimated asset-demand functions.
- **Main empirical innovation.** The paper estimates a *global asset demand system* using cross-country portfolio holdings, exchange rates, and asset prices, allowing substitution both *across countries within an asset class* and *across asset classes*.
- **Identification idea.** Use exogenous variation in *residual supply*: countries that are small but close to large investor blocs face stronger exogenous demand pressure through gravity, which shifts prices and exchange rates in a way that identifies demand elasticities.
- **Data.** Annual international portfolio holdings for 37 countries from 2003–2020, split into short-term debt, long-term debt, and equity.
- **Headline estimates.** Mean demand elasticities are **27.9** for short-term debt, **3.2** for long-term debt, and **1.2** for equity. Equity demand is therefore quite inelastic at the country level.
- **Variance decomposition.** Latent demand explains most exchange-rate and equity-price variation: about **82%** of exchange-rate changes and **60%** of market-to-book equity variation. By contrast, long-term yields are explained mostly by portfolio flows and macro variables.
- **Event-study application.** In the European sovereign debt crisis, Greece’s yield spike is mostly tied to macro deterioration, while Italy and Portugal are explained mainly by latent demand, especially from European investors.
- **US special status.** The paper estimates convenience yields of roughly **1.41%** on the US dollar, **2.71%** on US long-term debt, and **0.50%** on US equity, in expected-return units.
- **Why the paper matters.** It gives a quantitative asset-demand foundation for international finance puzzles such as exchange-rate disconnect, safe-asset demand, and the role of investor clientele in global asset pricing.

2 Data and measurement (key objects)

2.1 International portfolio holdings

- **Coverage.** 37 issuer countries, 37 investor countries, foreign-exchange reserves, and the rest of the world, from 2003 to 2020.
- **Asset classes.** Short-term debt, long-term debt, and equity.
- **Key adjustment.** Holdings are restated from issuer *residency* to *nationality* accounting using the Global Capital Allocation Project, to account for tax havens and offshore issuance.
- **Central banks.** Foreign-exchange reserves are treated as a separate investor group because the IMF aggregates central-bank holdings for confidentiality.

- **Outside asset.** The outside asset for each class includes assets issued outside the 37-country sample and foreign-currency debt issued by one of the 37 countries.
- **Why it matters.** The paper wants a demand system that exactly matches who holds what, by country and asset class.

2.2 Asset prices and macro characteristics

- **Exchange rates.** Year-end nominal exchange rates in US dollars per local-currency unit.
- **Real exchange rate.** Nominal exchange rate divided by the PPP conversion factor for GDP.
- **Short-term rate.** Three-month interbank rate.
- **Long-term rate.** Five-year zero-coupon yield, used as the representative long-term debt price object.
- **Equity price.** Market-to-book equity for the MSCI country index.
- **Country characteristics.** Log GDP, log GDP per capita, inflation, equity volatility, sovereign rating, bilateral distance, domestic-ownership indicator, and year fixed effects.

2.3 What Table 1 and Table 2 show

- **Home bias is huge.** In 2020, domestic investors held **92%** of US short-term debt, **84%** of US long-term debt, and **87%** of US equity.
- **Reserves matter for debt.** Foreign central banks held meaningful shares of developed-country debt, but little equity.
- **Large investors.** The United States, Japan, China, and reserves dominate the top-investor lists by asset class.
- **Interpretation.** The raw data already suggest gravity, home bias, and large investor-clientele effects.

2.4 Reduced-form price-quantity facts

- **Figure 1.** Relative short-term debt prices and quantities have a negative slope: countries with more relative supply have lower relative prices.
- **Figure 2.** The same negative relation appears for long-term debt in Germany, Japan, Switzerland, and the United Kingdom.
- **Figure 3.** Relative equity price and quantity also show a broadly downward-sloping relation, though less cleanly than debt.
- **Interpretation.** Assets across countries are not close to perfect substitutes. But these scatterplots are only suggestive because demand curves can shift.

3 Models and empirical specifications (core equations + interpretation)

3.1 Expected returns

$$r_{t+1}(n, l) + \Delta e_{t+1}(n) = -\theta_l p_t(n, l) - \Theta_l(e_t(n) - v_t(n)) + \iota_{n,l} + \nu_{t+1}(n, l). \quad (1)$$

$$\mathbb{E}_t[r_{t+1}(n, l) + \Delta e_{t+1}(n) - r_{t+1}(i, S) - \Delta e_{t+1}(i)] = \mu_{i,t}(n, l) + \iota_{n,l} - \iota_{i,S}, \quad (2)$$

$$\mu_{i,t}(n, l) = -\theta_l p_t(n, l) - \Theta_l(e_t(n) - v_t(n)) + \theta_S p_t(i, S) + \Theta_S(e_t(i) - v_t(i)). \quad (3)$$

- **Meaning.** Expected return in US dollars is forecast from the asset price and the real exchange rate.
- **Economic intuition.** A high price or a strong real exchange rate should predict lower future returns because of mean reversion.
- **Investor-specific excess return.** The relevant return for investor i is measured relative to investor i 's short-term domestic asset.
- **Interpretation.** Country allocation is driven by a *single index* in price and real exchange rate, rather than by two separate unrestricted objects.

3.2 Model of short-term rates

$$p_t(n, S) = \Pi' z_t(n) + \pi_t(n). \quad (4)$$

- **Meaning.** Short-term debt prices are a function of observable macro variables plus latent demand $\pi_t(n)$.
- **Interpretation.** This replaces the consumption Euler equation with a reduced-form interest-rate block.

3.3 Nested logit demand system

$$w_{i,t}(n, l) = w_{i,t}(n \mid l) w_{i,t}(l). \quad (5)$$

$$A_{i,t} = O_{i,t} + \sum_{l \in \{S, L, E\}} \sum_{n=1}^N E_t(n) P_t(n, l) Q_{i,t}(n, l). \quad (6)$$

$$w_{i,t}(n \mid l) = \frac{\delta_{i,t}(n, l)}{1 + \sum_{m=1}^N \delta_{i,t}(m, l)}, \quad (7)$$

$$\delta_{i,t}(n, l) = \exp(\lambda_l \mu_{i,t}(n, l) + \Lambda'_l x_{i,t}(n) + \epsilon_{i,t}(n, l)), \quad (8)$$

$$w_{i,t}(0 \mid l) = \frac{1}{1 + \sum_{n=1}^N \delta_{i,t}(n, l)}, \quad (9)$$

$$w_{i,t}(l) = \frac{\left(1 + \sum_{n=1}^N \delta_{i,t}(n, l)\right)^{\rho_l} \exp(\alpha_l + \xi_{i,t}(l))}{\sum_{k \in \{S, L, E\}} \left(1 + \sum_{n=1}^N \delta_{i,t}(n, k)\right)^{\rho_k} \exp(\alpha_k + \xi_{i,t}(k))}. \quad (10)$$

- **Inner nest.** Equation (7) allocates within an asset class across countries.
- **Outer nest.** Equation (10) allocates across short-term debt, long-term debt, and equity.
- **Demand shifters.** $\mu_{i,t}(n, l)$ captures expected return; $x_{i,t}(n)$ captures macro variables, bilateral distance, and home bias; $\epsilon_{i,t}(n, l)$ is latent demand within asset classes; $\xi_{i,t}(l)$ is latent demand across asset classes.
- **Role of ρ_l .** These parameters govern substitution across asset classes. If $\rho_l = 0$, no cross-asset substitution; if $\rho_l = 1$, the model collapses toward a logit with perfect cross-asset substitution.
- **Economic point.** The paper lets investors be sticky both across countries and across classes, which is essential for asset-price movements in general equilibrium.

3.4 Portfolio flows and market clearing

$$A_{i,t+1} = O_{i,t+1} + \sum_{l \in \{S, L, E\}} \sum_{n=1}^N (A_{i,t} w_{i,t}(n, l) + F_{i,t}(n, l)) \frac{E_{t+1}(n) P_{t+1}(n, l)}{E_t(n) P_t(n, l)}. \quad (13)$$

$$E_t(n) P_t(n, l) Q_t(n, l) = \sum_{i=1}^I A_{i,t} w_{i,t}(n, l). \quad (14)$$

$$[e'_t, p'_t]' = m(Q_t, X_t, O_t, \pi_t, \epsilon_t, \xi_t). \quad (15)$$

- **Portfolio flows.** $F_{i,t}(n, l)$ are active holdings changes excluding capital gains.
- **Market clearing.** Asset prices and exchange rates jointly adjust so that supply equals world demand across all countries and all three asset classes.
- **Equilibrium view.** Exchange rates and asset prices are equilibrium objects pinned down by holdings, characteristics, outside wealth, and latent demand.

3.5 Estimating equation for demand within asset class

$$\log \left(\frac{w_{i,t}(n | l)}{w_{i,t}(0 | l)} \right) = \lambda_l \mu_{i,t}(n, l) + \Lambda'_l x_{i,t}(n) + \epsilon_{i,t}(n, l). \quad (16)$$

- **Dependent variable.** Log portfolio weight on country n relative to the outside asset within asset class l .
- **Coefficient of interest.** λ_l is the demand elasticity parameter within asset class l .
- **Interpretation.** Countries with higher expected returns should receive larger portfolio weights, but the response is limited by imperfect substitution.

3.6 Reduced-supply identification example

$$\log\left(\frac{w_i(n)}{w_i(0)}\right) = -\lambda p(n) + \Lambda_1 D_i(n) + \Lambda_2 + \epsilon_i(n), \quad (17)$$

$$P(n)Q(n) = \sum_{i \in \{U, G\}} O_i \frac{w_i(n)}{w_i(0)}, \quad (18)$$

$$p(n) = m(Q(n), D_U(n), D_G(n), O, \epsilon(n)). \quad (19)$$

$$\text{Cov}(\epsilon_i(n), D_j(n) \mid D_i(n)) = 0, \quad (20)$$

$$\text{Cov}(p(n), D_j(n) \mid D_i(n)) \neq 0. \quad (21)$$

- **Figure 4 intuition.** US investors face lower residual supply of Dutch debt than Australian debt because German investors have stronger gravity-driven demand for Dutch debt.
- **Instrument logic.** Other investors' gravity-driven demand shifts residual supply faced by investor i without directly entering investor i 's demand equation.

3.7 Constructed instrument for inner-nest demand

$$IV_{i,t}(n, l) = \log\left(\sum_{j \neq i} \frac{O_{j,t} \hat{w}_{j,t}(n, l)}{1 - \sum_{l \in \{S, L, E\}} \sum_{m=1}^N \hat{w}_{j,t}(m, l)}\right) - (v_t(n) + \hat{q}_t(n, l)). \quad (28)$$

- **First term.** Predicted log demand from all other investors, built only from inelastic time-invariant components such as bilateral distance and outside wealth.
- **Second term.** Predicted log supply, built from country size variables such as GDP and population.
- **Interpretation.** The IV is a residual-supply shifter. Small countries near large investor blocs should have high demand pressure and therefore low residual supply.

3.8 Demand across asset classes

$$\log\left(\frac{w_{i,t}(l)}{w_{i,t}(E)}\right) = -\rho_l \log(w_{i,t}(0 \mid l)) + \rho_E \log(w_{i,t}(0 \mid E)) + \alpha_l + \xi_{i,t}(l). \quad (29)$$

$$\hat{w}_{i,t}(0 \mid l) = \frac{1}{1 + \sum_{n=1}^N \hat{\delta}_i(n, l)}. \quad (33)$$

$$\mathbb{E}[\xi_{i,t} \mid \Upsilon_{i,S}, \Upsilon_{i,L}, \Upsilon_{i,E}] = 0. \quad (34)$$

- **Idea.** Variation in investor-specific home bias shifts the outside portfolio weight and identifies substitution across asset classes.
- **Economic interpretation.** If an investor has very strong home bias in long-term debt, then the outside weight in long-term debt is low, which identifies how tightly long-term debt competes with short-term debt and equity.

3.9 Aggregate demand elasticity and variance decomposition

$$-\frac{\partial \log \left(\sum_{i=1}^I Q_{i,t}(n, l) \right)}{\partial p_t(n, l)} = 1 - \frac{\partial \log \left(\sum_{i=1}^I A_{i,t} w_{i,t}(n, l) \right)}{\partial p_t(n, l)}. \quad (35)$$

$$\Delta e_{t+1} = (e(Q_{t+1}) - e_t) + (e(X_{t+1}) - e(Q_{t+1})) + (e_{t+1} - e(X_{t+1})). \quad (36)$$

$$1 = \frac{\text{Cov}(e(Q_{t+1}) - e_t, \Delta e_{t+1})}{\text{Var}(\Delta e_{t+1})} + \frac{\text{Cov}(e(X_{t+1}) - e(Q_{t+1}), \Delta e_{t+1})}{\text{Var}(\Delta e_{t+1})} + \frac{\text{Cov}(e_{t+1} - e(X_{t+1}), \Delta e_{t+1})}{\text{Var}(\Delta e_{t+1})}. \quad (37)$$

- **Equation (35).** This converts the estimated demand system into an economically interpretable price elasticity of country-level demand.
- **Equations (36)–(37).** These decompose exchange-rate movements into portfolio flows, macro demand shifters, and latent-demand shocks.
- **Interpretation.** The exchange rate disconnect puzzle becomes a decomposition statement: macro observables explain only a small share of exchange-rate variation compared with latent demand.

4 Identification and instruments (what makes the estimates credible)

4.1 Inner nest: residual supply from geography and size

- The first identification problem is endogeneity of expected returns and prices in equation (16).
- The paper isolates an **inelastic and time-invariant** component of demand using bilateral distance and an inelastic component of supply using GDP and population.
- This generates an instrument equal to **predicted demand from everyone except investor i minus predicted supply**.
- The gravity effect is essential: countries near large investor blocs receive higher exogenous demand pressure.
- Figure 5 is the key visual: the Netherlands and Australia are similarly sized on the supply axis, but Dutch debt faces much larger predicted demand because of European proximity.
- First-stage F-statistics are very strong: roughly **130** for short-term debt, **1,297** for long-term debt, and **521** for equity.

4.2 Outer nest: home bias as a shifter of outside weights

- The second identification problem is endogeneity of the outside portfolio weights in equation (29).
- The paper uses investor-specific home bias to generate variation in the outside portfolio weights.

- Investors with stronger home bias toward domestic debt mechanically place smaller weight on outside debt, shifting the inclusive value and identifying cross-asset substitution.
- The weak-IV F-statistic is **404**, so the outer-nest instruments are also very strong.

4.3 Why the design is compelling

- It uses **cross-sectional** variation in geography and market size, not short-run time-series events.
- It gives a direct estimate of asset demand elasticities rather than backing them out from aggregate moments.
- It separately identifies within-class and across-class substitution, which is crucial for exchange rates and long-term yields.

4.4 Main caveats

- Country-level portfolio data may hide heterogeneity across institutions within a country.
- The outside asset bundles many things together, including foreign-currency debt and assets outside the 37-country sample.
- The demand-system interpretation of the variance decomposition is structural only under strong exogeneity assumptions on the state variables.
- Convenience-yield estimates are upward biased if omitted US characteristics also attract extra demand.

5 Tables and figures

5.1 Table 1: Market values of financial assets

Read:

- The table shows the sheer size of US markets and the extent of domestic ownership. This is the empirical basis for home bias.
- Foreign-exchange reserves hold a meaningful share of developed-country debt, especially in Europe and the United States.
- If the data looked globally frictionless, the whole demand-system approach would be less compelling. They do not.

5.2 Table 2: Top ten investors by asset class

Read:

- The largest investors are the United States, Japan, China, and reserve managers.
- Reserves rank among the largest investors in debt markets, which foreshadows their importance in exchange-rate and convenience-yield decompositions.

TABLE 1
MARKET VALUES OF FINANCIAL ASSETS

Issuer	Short-term debt			Long-term debt			Equity		
	Billion US\$	Domestic share	Share in reserves	Billion US\$	Domestic share	Share in reserves	Billion US\$	Domestic share	Share in reserves
<i>Developed markets: North America</i>									
Canada	521	0.89	0.06	2,522	0.77	0.05	6,515	0.87	0.00
United States	5,489	0.92	0.04	41,070	0.84	0.05	55,623	0.87	0.00
<i>Developed markets: Europe</i>									
Austria	18	0.15	0.65	559	0.49	0.08	282	0.79	0.00
Belgium	121	0.69	0.11	1,152	0.62	0.06	1,739	0.93	0.00
Finland	69	0.60	0.16	384	0.43	0.09	678	0.78	0.00
France	628	0.79	0.12	5,065	0.65	0.07	10,614	0.89	0.00
Germany	321	0.43	0.34	4,342	0.58	0.13	3,672	0.57	0.00
Italy	168	0.62	0.01	3,749	0.78	0.00	2,384	0.89	0.00
Netherlands	67	0.40	0.25	1,235	0.45	0.08	6,057	0.80	0.00
Portugal	26	0.60	0.01	336	0.69	0.00	386	0.92	0.00
Spain	195	0.65	0.12	2,621	0.66	0.04	2,229	0.88	0.00
Denmark	48	0.58	0.29	670	0.75	0.02	1,705	0.85	0.00
Israel	29	0.90	0.01	509	0.90	0.00	660	0.86	0.00
Norway	30	0.60	0.08	224	0.19	0.11	1,038	0.91	0.00
Sweden	123	0.55	0.07	428	0.46	0.05	3,019	0.88	0.00
Switzerland	115	0.76	0.16	908	0.88	0.03	3,747	0.75	0.00
United Kingdom	341	0.84	0.04	4,163	0.84	0.04	6,666	0.76	0.00
<i>Developed markets: Pacific</i>									
Australia	202	0.89	0.04	1,824	0.71	0.04	1,763	0.73	0.00
Hong Kong	13	0.40	0.06	55	0.11	0.02	2,454	0.81	0.00
Japan	1,889	0.74	0.12	13,467	0.96	0.01	12,172	0.87	0.00
New Zealand	9	0.71	0.10	125	0.76	0.02	1,030	0.96	0.00
Singapore	41	0.04	0.39	225	0.66	0.03	887	0.82	0.00
<i>Emerging markets</i>									
Greece	16	0.71	0.00	134	0.86	0.01	182	0.91	0.00
Brazil	399	0.98	0.00	1,434	0.90	0.00	2,399	0.91	0.00
China	455	0.80	0.01	17,359	0.97	0.00	15,002	0.77	0.00
Colombia	27	0.99	0.00	185	0.81	0.00	483	0.99	0.00
Czech Republic	28	0.00	0.00	95	0.54	0.00	106	0.98	0.00
Hungary	7	0.99	0.00	107	0.79	0.00	104	0.89	0.00
India	326	0.98	0.00	2,027	0.97	0.00	2,354	0.80	0.00
Malaysia	18	0.83	0.01	342	0.84	0.00	447	0.89	0.00
Mexico	94	0.96	0.00	591	0.78	0.00	1,313	0.94	0.00
Philippines	17	0.99	0.00	94	0.78	0.00	261	0.88	0.00
Poland	42	0.99	0.00	301	0.77	0.00	255	0.91	0.00
Russia	9	0.42	0.10	298	0.69	0.01	3,039	0.96	0.00
South Africa	45	0.98	0.01	196	0.75	0.01	1,105	0.91	0.00
South Korea	309	0.92	0.03	2,103	0.92	0.02	3,147	0.85	0.00
Thailand	75	0.98	0.00	338	0.92	0.00	538	0.86	0.00

Note.—This table reports only local currency debt. All market values are in billion US dollars at year-end 2020.

TABLE 2
TOP TEN INVESTORS BY ASSET CLASS

Short-term debt		Long-term debt		Equity	
Investor	Billion US\$	Investor	Billion US\$	Investor	Billion US\$
United States	5,423	United States	38,283	United States	56,324
Japan	1,444	China	17,331	Japan	12,424
Reserves	1,025	Japan	16,206	China	11,952
France	827	United Kingdom	5,752	France	10,376
United Kingdom	496	Germany	5,513	Canada	7,361
Canada	471	France	5,490	United Kingdom	6,800
China	440	Reserves	4,952	Netherlands	5,971
Brazil	395	Italy	3,721	Germany	3,393
India	325	Canada	2,979	Switzerland	3,390
South Korea	301	South Korea	2,350	Hong Kong	3,240

Note.—The International Monetary Fund (2003–2020a) aggregates foreign exchange reserves across all foreign central banks for confidentiality. All market values are in billion US dollars at year-end 2020.

- The table highlights that the relevant decision makers are not just private investors; official institutions matter.

5.3 Figures 1–3: Relative prices versus quantities

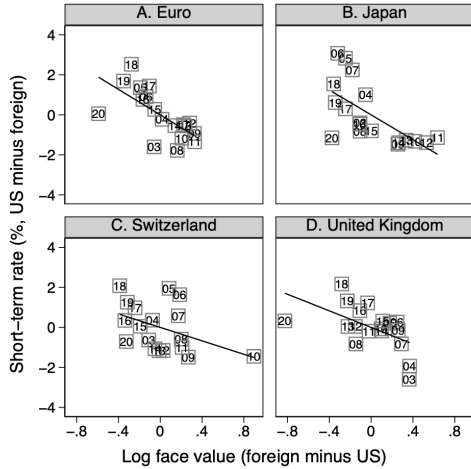


Figure 1. Relative short-term debt prices versus quantities. The vertical axis is the US short-term rate minus each region's short-term rate, reported in annual percentage points. The horizontal axis is each region's log face value of short-term debt in US dollars minus log face value of US short-term debt. Each axis is demeaned by the time-series mean. The two-digit number represents year (e.g., 03 is 2003). Each panel shows the linear regression line.

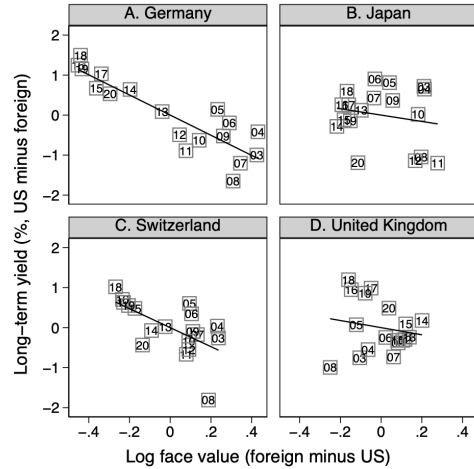


Figure 2. Relative long-term debt prices versus quantities. The vertical axis is the US long-term yield minus each country's long-term yield, reported in annual percentage points. The horizontal axis is each country's log face value of long-term debt in US dollars minus log face value of US long-term debt. Each axis is demeaned by the time-series mean. The two-digit number represents year (e.g., 03 is 2003). Each panel shows the linear regression line.

Read:

- Short-term debt, long-term debt, and equity all show broadly downward-sloping relative demand curves across countries.
- Debt evidence is especially clean: the same countries repeatedly trace out a negative price-

quantity relation relative to the United States.

- These figures are only motivation, not identification. The main point is that cross-country assets are imperfect substitutes.

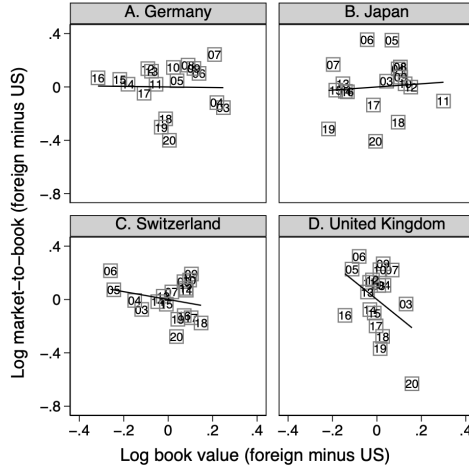


Figure 3. Relative equity prices versus quantities. The vertical axis is each country's log market-to-book equity minus the US log market-to-book equity. The horizontal axis is each country's log book equity in US dollars minus the US log book equity. Each axis is demeaned by the time-series mean. The two-digit number represents year (e.g., 03 is 2003). Each panel shows the linear regression line.

TABLE 3
PREDICTIVE REGRESSIONS

Variable	Exchange rate	Long-term debt	Equity
Log asset price		-0.74 (0.11)	-0.15 (0.22)
Log real exchange rate	-0.27 (0.07)	-0.36 (0.07)	-0.54 (0.28)
Constant		-0.07 (0.02)	0.25 (0.20)
R^2	0.17	0.32	0.12
Observations	424	640	640

Note.—Log asset price is minus maturity times log yield for long-term debt and log market-to-book for equity. All models include country fixed effects. Robust standard errors clustered by year are reported in parentheses. The annual sample period is 2003 to 2020.

5.4 Table 3: Predictive regressions

Read:

- For long-term debt, a higher price predicts lower future returns, with a price coefficient around -0.74 , and a stronger real exchange rate predicts lower returns with coefficient -0.36 .
- For equity, the corresponding coefficients are about -0.15 on market-to-book and -0.54 on the real exchange rate.
- These regressions supply the expected-return index that enters asset demand. They are economically sensible and statistically informative.

5.5 Figure 4: Illustration of residual supply

Read:

- German investors want Dutch debt more than Australian debt because of geography.
- Hence US investors face less residual supply of Dutch debt even though Dutch and Australian supply are similar in size.
- This picture is the conceptual core of the paper's IV strategy.

5.6 Table 4: Estimated demand within asset class

Read:

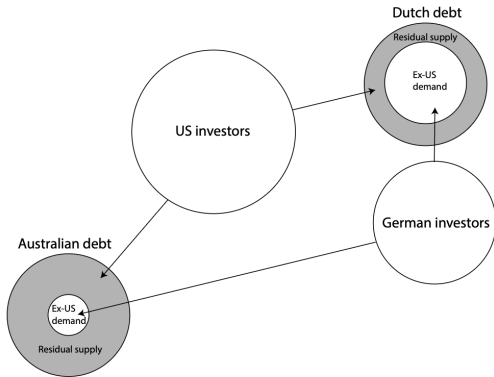


Figure 4. Illustration of residual supply. This figure illustrates the residual supply of Dutch and Australian debt from the perspective of US investors. German investors have a higher demand for Dutch than Australian debt through a gravity effect that depends on the bilateral distance between countries.

TABLE 4
ESTIMATED DEMAND WITHIN ASSET CLASS

Variable	Short-term debt	Long-term debt	Equity
Expected return	14.33 (2.32)	4.52 (0.51)	10.33 (0.79)
Log GDP	1.28 (0.02)	1.10 (0.01)	1.32 (0.02)
Log GDP per capita	3.67 (0.35)	2.16 (0.11)	3.68 (0.19)
Inflation	-23.49 (4.22)	-9.22 (1.79)	-16.56 (1.88)
Volatility	-2.83 (0.40)	-0.52 (0.27)	-5.89 (0.36)
Rating	-0.77 (1.26)	10.24 (1.29)	13.96 (1.23)
Distance	-0.08 (0.01)	-0.18 (0.00)	-0.15 (0.01)
Indicator variables:			
Domestic ownership	8.46 (0.18)	6.19 (0.09)	7.69 (0.14)
Reserves	0.01 (0.19)	0.10 (0.10)	-2.83 (0.14)
Other countries	0.78 (0.17)	0.77 (0.06)	-1.86 (0.10)
Constant	-52.35 (3.67)	-34.78 (1.15)	-50.94 (2.14)
<i>F</i> -statistic for weak IV	130	1,297	521
Observations	20,549	23,431	23,779

Note.—Expected returns are the predicted values from the predictive regressions in Table 3. The sovereign debt rating is a continuous measure equal to -1 times the ten-year default rate. All models include year fixed effects. Heteroskedasticity-robust standard errors are reported in parentheses. The critical value for a test of weak instruments at the 5 percent significance level is 16.38 (Stock and Yogo 2005, Table 5.2). The annual sample period is 2003 to 2020.

- Expected-return coefficients are large and positive: 14.33 for short-term debt, 4.52 for long-term debt, and 10.33 for equity.
- Bilateral distance enters strongly negatively and home bias enters strongly positively, confirming gravity and domestic-preference patterns.
- The weak-IV statistics are huge, so the identification is not fragile.
- The economic message is that investors chase expected returns, but they do so within a world of substantial gravity and home-bias frictions.

5.7 Table 5: Estimated demand across asset classes

Read:

- The outer-nest elasticities are $\rho_S = 0.30$, $\rho_L = 0.61$, and $\rho_E = 0.62$.
- Substitution across asset classes is therefore important and especially strong for long-term debt and equity.
- This matters because shocks to one asset class can spill into the demand for others through the inclusive value.

Variable	Symbol	Estimate
Log outside portfolio weight:		
Short-term debt	ρ_S	0.30 (0.03)
Long-term debt	ρ_L	0.61 (0.06)
Equity	ρ_E	0.62 (0.04)
Indicator variables:		
Short-term debt	α_S	-0.82 (0.20)
Long-term debt	α_L	1.13 (0.22)
F -statistic for weak IV		404
Observations		1,352

Note.—Heteroskedasticity-robust standard errors are reported in parentheses. The annual sample period is 2003 to 2020.

TABLE 6
ESTIMATED MODEL OF
SHORT-TERM RATES

Variable	Coefficient
Log GDP	0.01 (0.01)
Log GDP per capita	-0.01 (0.01)
Inflation	-0.11 (0.01)
Volatility	0.00 (0.01)
Rating	-0.03 (0.03)
Constant	-0.05 (0.03)
Observations	442

Note.—The model includes country fixed effects. Heteroskedasticity-robust standard errors are reported in parentheses. The annual sample period is 2003 to 2020.

TABLE 7
VARIANCE DECOMPOSITION OF EXCHANGE RATES AND ASSET PRICES

Change in	Exchange rate	Short-term rate	Long-term yield	Market-to-book equity
Portfolio flows	0.02 (0.05)		0.54 (0.18)	0.19 (0.07)
Macro variables	0.16 (0.06)	0.14 (0.06)	0.43 (0.10)	0.21 (0.03)
Latent demand	0.82 (0.07)	0.86 (0.06)	0.03 (0.26)	0.60 (0.07)
Reserves	0.10 (0.01)		0.02 (0.03)	-0.01 (0.01)
North America	0.32 (0.09)	0.45 (0.14)	-0.21 (0.19)	0.13 (0.07)
Europe	0.21 (0.04)	0.17 (0.06)	0.16 (0.06)	0.26 (0.04)
Pacific	0.22 (0.06)	0.03 (0.02)	-0.02 (0.02)	0.09 (0.02)
Emerging markets	-0.05 (0.02)	0.21 (0.06)	0.09 (0.07)	0.12 (0.05)
Other countries	0.02 (0.01)		0.00 (0.01)	0.00 (0.00)
Observations	399	416	603	603

Note.—Heteroskedasticity-robust standard errors are reported in parentheses. The observations are value-weighted by the market weights within year and asset class. The annual sample period is 2003 to 2020.

5.8 Table 6: Estimated model of short-term rates

Read:

- Inflation is the dominant macro shifter: the coefficient is about -0.11 on the short-term debt price.
- GDP, GDP per capita, volatility, and rating are much smaller in magnitude.
- Thus, the short-rate block is mainly a reduced-form inflation channel plus latent demand.

5.9 Table 7: Variance decomposition of exchange rates and asset prices

Read:

- Exchange rates: portfolio flows explain almost nothing (2%), macro variables explain 16%, and latent demand explains 82%.
- Short-term rates: macro variables explain only 14%; latent demand explains 86%.
- Long-term yields: portfolio flows explain 54%, macro variables 43%, and latent demand only 3% on average.
- Equity prices: portfolio flows explain 19%, macro variables 21%, and latent demand 60%.
- This is the empirical heart of the paper: exchange rates and equity valuations are mostly about who wants to hold what, not only about macro fundamentals.

5.10 Table 8 and Figure 6: The European sovereign debt crisis

Read:

- Greece: macro variables explain 46% of the long-term yield variation, latent demand 32%.

TABLE 8
VARIANCE DECOMPOSITION OF LONG-TERM
YIELDS IN THE EURO AREA

Change in	Greece	Italy	Portugal
Portfolio flows	0.22 (0.11)	0.14 (0.17)	0.24 (0.04)
Macro variables	0.46 (0.09)	-0.28 (0.17)	0.02 (0.19)
Latent demand	0.32 (0.03)	1.15 (0.31)	0.74 (0.21)
Reserves	0.00 (0.01)	0.02 (0.09)	-0.01 (0.03)
North America	0.01 (0.01)	0.08 (0.07)	0.05 (0.03)
Europe	0.19 (0.01)	0.98 (0.19)	0.65 (0.16)
Pacific	0.02 (0.00)	0.05 (0.02)	0.05 (0.01)
Emerging markets	0.08 (0.02)	0.00 (0.01)	0.00 (0.00)
Other countries	0.02 (0.01)	0.01 (0.00)	0.01 (0.00)
Observations	17	17	17

Note.—Heteroskedasticity-robust standard errors are reported in parentheses. The annual sample period is 2003 to 2020.

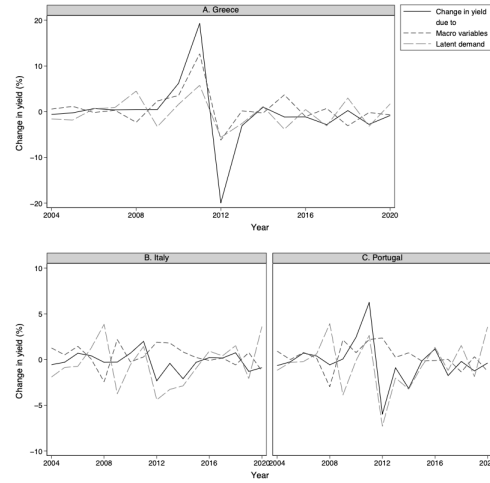


Figure 6. Long-term yields in the euro area. Annual changes in the long-term yields are decomposed into portfolio flows and shifts in asset demand through macro variables and latent demand. This figure reports the changes due to the macro variables and latent demand only.

- Italy: latent demand explains about 115% of yield variation, while macro variables contribute negatively on net.
- Portugal: latent demand explains 74% of yield variation.
- European investors alone explain 98% of Italian-yield variation and 65% of Portuguese-yield variation.
- Interpretation: Greece looks like realized insolvency; Italy and Portugal look like market-perceived vulnerability and demand-driven panic.

5.11 Table 9: Mean latent demand

Read:

- Foreign investors' latent demand for US assets is uniformly positive across years and asset classes.
- The corresponding euro-area latent demand is not uniformly positive and deteriorates after the sovereign-debt crisis.
- This is the evidence that US assets enjoy special status over and above measured characteristics.

5.12 Table 10 and Figure 7: Convenience yield on US assets

Read:

- Without special demand for US assets, the dollar would be weaker by about 5.23%, implying a dollar convenience yield of 1.41% in expected-return units.

TABLE 9
MEAN LATENT DEMAND

Year	United States			Euro area		
	Short-term debt	Long-term debt	Equity	Short-term debt	Long-term debt	Equity
2003	2.6	2.0	2.6	4.5	2.3	4.0
2004	1.5	1.6	1.7	4.3	2.0	3.7
2005	2.1	1.4	2.0	3.9	1.4	2.9
2006	1.3	1.4	1.3	3.3	1.4	3.0
2007	1.1	1.2	1.0	3.0	1.1	3.1
2008	1.8	1.4	1.4	4.4	1.5	3.4
2009	1.3	0.9	0.8	3.5	1.1	2.5
2010	1.5	0.8	0.7	3.0	0.9	1.7
2011	1.2	1.2	0.7	2.5	0.3	1.1
2012	0.4	0.5	-0.1	2.3	0.4	0.9
2013	0.5	0.2	0.2	1.2	0.2	0.6
2014	0.6	0.3	0.3	1.0	0.1	-0.4
2015	0.8	0.5	0.9	0.8	0.3	0.1
2016	1.0	0.6	0.8	0.4	-0.3	-0.6
2017	0.3	0.3	0.3	0.1	-0.1	-0.7
2018	0.8	0.5	1.1	0.3	-0.1	-0.8
2019	0.4	0.0	0.9	-0.3	-0.3	-1.6
2020	0.5	0.4	2.1	0.2	0.1	0.5
Mean	1.1	0.8	1.0	2.0	0.6	1.2

Note.—This table reports the cross-sectional mean of latent demand by year, issuer region, and asset class, excluding the domestic investors' latent demand. Latent demand includes the year fixed effects from the asset demand estimation. The last row reports the overall mean.

TABLE 10
CONVENIENCE YIELD ON US ASSETS

Investor	Foreign short-term debt		US long-term debt		US equity	
	Exchange rate	Expected return	Yield	Expected return	Market-to-book	Expected return
Total	5.23	-1.41	0.73	2.71	-3.35	0.50
	(0.56)	(0.15)	(0.09)	(0.35)	(0.39)	(0.06)
Reserves	3.41	-0.92	0.28	1.03	-0.08	0.01
	(0.43)	(0.12)	(0.03)	(0.13)	(0.02)	(0.00)
North America	0.06	-0.02	0.01	0.05	-0.37	0.05
	(0.01)	(0.00)	(0.00)	(0.00)	(0.04)	(0.01)
Europe	0.84	-0.23	0.21	0.80	-1.75	0.26
	(0.09)	(0.02)	(0.03)	(0.11)	(0.20)	(0.03)
Pacific	0.52	-0.14	0.20	0.73	-0.87	0.13
	(0.07)	(0.02)	(0.03)	(0.10)	(0.10)	(0.01)
Emerging markets	0.13	-0.04	0.01	0.05	-0.16	0.02
	(0.02)	(0.00)	(0.00)	(0.01)	(0.04)	(0.01)
Other countries	0.26	-0.07	0.01	0.05	-0.12	0.02
	(0.03)	(0.01)	(0.00)	(0.01)	(0.02)	(0.00)

Note.—This table reports the time-series mean of the counterfactual changes in exchange rates and asset prices in the absence of special demand for US assets, reported in annual percentage points. Special demand is estimated as the cross-sectional mean of latent demand for US assets by year and asset class, excluding the US investors' latent demand. Expected returns are the predicted values from the predictive regressions in Table 3, based on the counterfactual asset prices and exchange rates. Heteroskedasticity-robust standard errors are reported in parentheses. The annual sample period is 2003 to 2020.

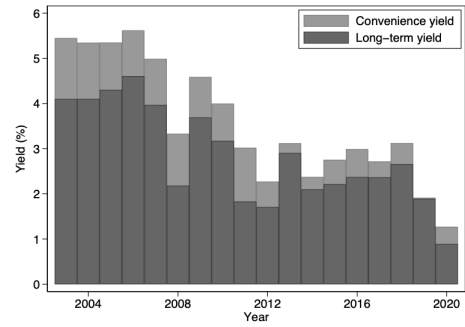


Figure 7. Convenience yield on US long-term debt. This figure reports the change in the long-term yield in the absence of special demand for US long-term debt, reported in annual percentage points. Special demand is estimated as the cross-sectional mean of latent demand for US long-term debt by year, excluding the US investors' latent demand.

- US long-term yields would be about 0.73 percentage points higher, corresponding to a convenience yield of 2.71%.
- US equity market-to-book would be lower by about 3.35%, implying a smaller equity convenience yield of 0.50%.
- Reserves and European investors are major contributors, especially for debt.
- Figure 7 shows that the long-bond convenience yield remains large, though somewhat lower in the low-rate environment.

6 Short summary

- **Conclusion.** A global asset-demand system with gravity, home bias, and latent investor demand explains a large share of exchange-rate and asset-price movements.
- **Main empirical lesson.** Country-level asset demand is inelastic, especially for equity and long-term debt, and latent demand is quantitatively central.
- **Why this matters.** To understand exchange rates, safe assets, and global imbalances, one must model who holds assets and how imperfectly substitutable those assets are.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** The paper estimates a tractable international asset-demand system that matches observed cross-country portfolios and shows that investor demand is a first-order determinant of global asset prices.
- **Magnitude.** Demand elasticities are very high for short-term debt, much smaller for long-term debt, and smallest for equity, implying substantial price pressure in global markets.
- **Credibility.** Identification comes from gravity-driven residual supply and home-bias-driven outside weights, with very strong first-stage evidence.
- **Mechanism evidence.** Latent demand, rather than observable macro variables, explains most exchange-rate and equity-price variation.
- **Event evidence.** The European sovereign debt crisis decomposes neatly into a Greece insolvency story and an Italy/Portugal investor-demand story.
- **Quantitative implication.** US asset convenience yields are large, especially for the dollar and long-term Treasuries.

7.2 Economic intuition

- Investors do not treat global assets as frictionless substitutes. Geography, home bias, and preferred habitats matter.
- When some investor groups shift their demand, prices and exchange rates must move to induce others to absorb the opposite side.

- Because these demand shifts are large and assets are imperfect substitutes, latent demand can dominate observable fundamentals in explaining asset-price fluctuations.

7.3 Takeaway

This paper is a bridge between international macro and asset pricing. Exchange rates, sovereign yields, and equity valuations are not pinned down only by fundamentals and arbitrage; they are also equilibrium outcomes of a world in which different investor clienteles want different assets for different reasons. The paper's main contribution is to measure those demand forces directly and show that they are quantitatively central.